ROOM CLIMATE MONITOR

EE108 Assignment — 02 by Team 04

13 November 2024

OUR TEAM

Md Rayhan Mia
PowerPoint / Code

Ales Vittek
TinkerCad / Code

Ava Brennan
TinkerCad / Code

Kris. Vitkauskas
Hardware Assembly / Code

Dharan Kumar
Hardware Assembly / Code

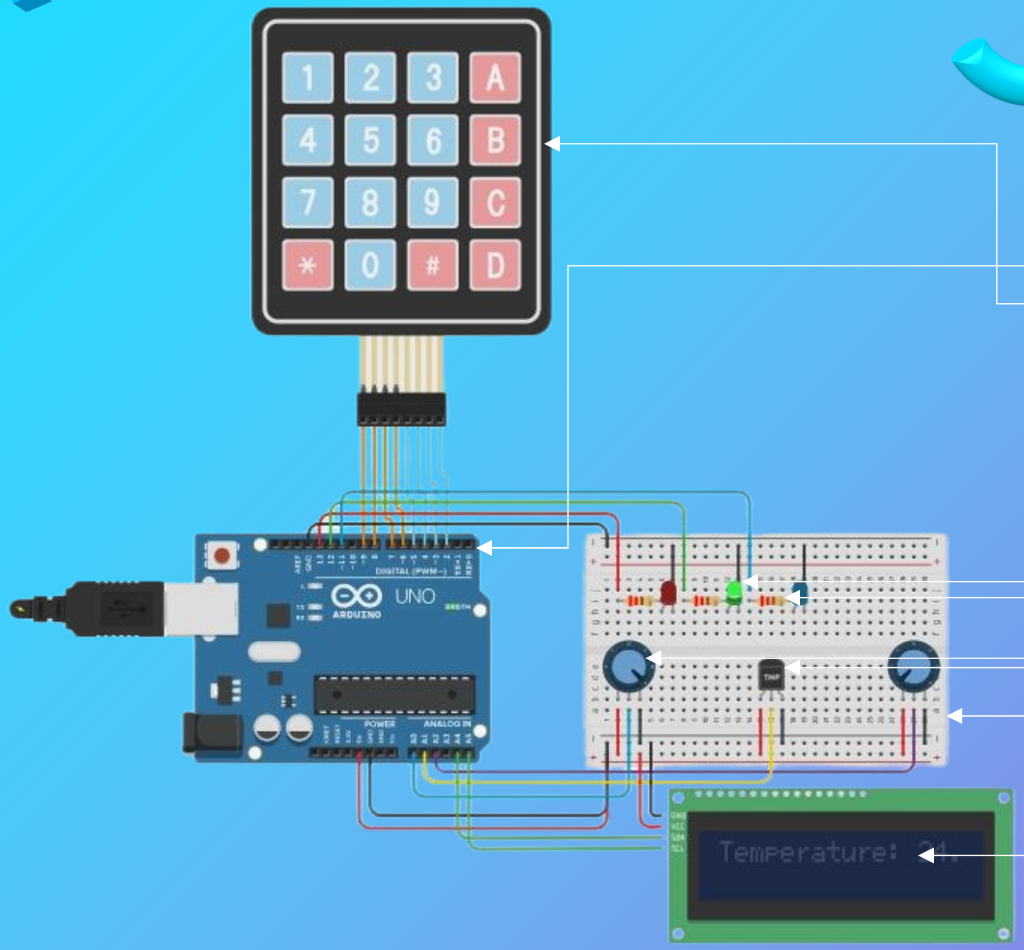
PROJECT OVERVIEW

Objective

To create a device that monitors room temperature, humidity, and pressure, displaying the data on an LCD and indicating status with LEDs.

Key Features

- User interaction via a 4x4 keypad.
- Real-time sensor readings displayed on an LCD.
- LED indicators (Red, Green, Blue) to signify sensor status.



Components Used

- **Arduino Uno:** The brain of the project.4x4
- Keypad: User input device.
- **LCD Monitor:** Display for sensor readings.
- **TMP36 Temperature Sensor:** Measures temperature.
- **Potentiometers:** Simulate humidity and pressure sensors in Tinkercad.
- **LEDs (Red, Green, Blue):** Visual indicators for sensor status.
- **Resistors:** Protect LEDs from excessive current.
- **Breadboard and Jumper Wires:** For circuit connections.

Code Part 1 – Libraries and Pin Definitions

```cpp
#include <Wire.h>
#include <Adafruit_LiquidCrystal.h>
#include <Keypad.h>

// Initialize LCD with I2C address
Adafruit_LiquidCrystal lcd(0);

// Keypad setup
const byte ROWS = 4;
const byte COLS = 4;
char keys[ROWS][COLS] = {
  {'1', '2', '3', 'A'},
  {'4', '5', '6', 'B'},
  {'7', '8', '9', 'C'},
  {'*', '0', '#', 'D'}
};
byte rowPins[ROWS] = {9, 8, 7, 6};
byte colPins[COLS] = {5, 4, 3, 2};
Keypad keypad = Keypad(makeKeymap(keys), rowPins, colPins, ROWS, COLS);

// Sensor and LED pins
const int humidityPin = A0;
const int tempPin = A1;
const int pressurePin = A2;
const int redLED = 13;
const int greenLED = 12;
const int blueLED = 11;
```

- **Wire Library:** Facilitates I2C communication for the LCD.
- **Adafruit LiquidCrystal Library:** Controls the I2C LCD display.
- **Keypad Library:** Manages input from the 4x4 keypad.
- **Initialization:** Setting up LCD, keypad, sensor, and LED pins as constants for easy reference.

Code Part 2 – Setup Function

- **LCD Initialization:** Sets up the display size and backlight.
- **LED Pin Configuration:** Sets the LED pins as outputs to control them.
- **Welcome Message:** Provides initial instructions to the user.

```
void setup() {
  Serial.begin(9600);

  // Initialize LCD
  lcd.begin(16, 2);
  lcd.setBacklight(LOW);
  lcd.clear();

  // Set LED pins as OUTPUT
  pinMode(redLED, OUTPUT);
  pinMode(greenLED, OUTPUT);
  pinMode(blueLED, OUTPUT);

  // Display welcome message
  lcd.setCursor(0, 0);
  lcd.print("Sensor Project");
  lcd.setCursor(0, 1);
  lcd.print("Press 1, 2 or 3...");

  delay(2000);
  lcd.clear();
}
```

```
void loop() {
  char key = keypad.getKey();  // Get keypress from the keypad
  if (key) {
    Serial.print("Key Pressed: ");
    Serial.println(key);  // Debugging: Print key to Serial Monitor
    if (key == '1' || key == '2' || key == '3') { // Process the keypress
      lcd.clear(); // Clear display for new output
      // Process the keypress
      if (key == '1'){displayHumidity();}
      else if (key == '2') {displayTemperature();}
      else if (key == '3') {displayPressure();
      }
      delay(2000);  // Wait before clearing
      lcd.clear();lcd.setCursor(0, 0);
      lcd.print("Press 1, 2 or 3...");
    } else {
      // Handle invalid keys
      lcd.clear();
      lcd.setCursor(0, 0);
      lcd.print("Invalid Key");
      delay(1000);
      lcd.clear();
      lcd.setCursor(0, 0);
      lcd.print("Press 1, 2 or 3...");
    }
  }
}
```

Code Part 3 – Keypad Input Handling

- **Key Detection:** Continuously checks for key presses.
- **Valid Keys:** Only '1', '2', and '3' trigger sensor readings.
- **Invalid Keys:** Display "Invalid Key" message for unassigned keys.

Code Part 4 – Sensor Reading Functions

- displayHumidity() - Reads humidity, maps to %, shows on LCD, sets LEDs.
- displayTemperature() - Reads temp, converts to °C, shows on LCD, sets LEDs.
- displayPressure() - Reads pressure, maps to 950-1050 hPa, shows on LCD, sets LEDs.

```
void displayHumidity() {
  int humidityValue = analogRead(humidityPin);
  float humidity = map(humidityValue, 0, 1023, 0, 100);  // Map to 0-100%
  // Display humidity on LCD
  lcd.setCursor(0, 0);lcd.print("Humidity: ");lcd.print(humidity);
  lcd.print(" %");
  // LED logic for humidity
  if (humidity > 80) {setLEDs(RED);} else if (humidity > 40) {setLEDs(GREEN);} else {setLEDs(BLUE);}}

void displayTemperature() {
  int tempValue = analogRead(tempPin);
  float voltage = tempValue * (5.0 / 1023.0);  // Convert to voltage
  float temperature = (voltage - 0.5) * 100;   // Convert to Celsius
  // Display temperature on LCD
  lcd.setCursor(0, 0);lcd.print("Temperature: ");
  lcd.print(temperature);lcd.print(" C");
  // LED logic for temperature
  if (temperature > 30) {setLEDs(RED);
  } else if (temperature > 20) {setLEDs(GREEN);} else {setLEDs(BLUE);}}

void displayPressure() {
  int pressureValue = analogRead(pressurePin);
  float pressure = map(pressureValue, 0, 1023, 950, 1050);  //950-1050hPa
  // Display pressure on LCD
  lcd.setCursor(0, 0);lcd.print("Pressure: ");
  lcd.print(pressure);lcd.print(" hPa");
  // LED logic for pressure
  if (pressure > 1020) {setLEDs(RED);
  } else if (pressure > 1000) {setLEDs(GREEN);} else {setLEDs(BLUE);}}
```

Code Part 5 – LED Control Function

```
// LED logic for humidity
        if (humidity > 80) {
          digitalWrite(blueLED, LOW); digitalWrite(greenLED, LOW);
          digitalWrite(redLED, HIGH); // Red LED for high humidity
        } else if (humidity > 40) {
          digitalWrite(blueLED, LOW); digitalWrite(redLED, LOW);
          digitalWrite(greenLED, HIGH); //Green LED for moderate Humidity
        } else {
          digitalWrite(redLED, LOW); digitalWrite(greenLED, LOW);
          digitalWrite(blueLED, HIGH);} // Blue LED for low humidity
// LED logic for temperature
        if (temperature > 30) {
          digitalWrite(blueLED, LOW); digitalWrite(greenLED, LOW);
          digitalWrite(redLED, HIGH); // Red LED for too high temperature
        } else if (temperature > 20) {
          digitalWrite(blueLED, LOW); digitalWrite(redLED, LOW);
          digitalWrite(greenLED, HIGH); // Green LED for normal temp
        } else {
          digitalWrite(redLED, LOW); digitalWrite(greenLED, LOW);
          digitalWrite(blueLED, HIGH);} // Blue LED for low temperature
// LED logic for pressure
        if (pressure > 1020) {
          digitalWrite(blueLED, LOW);digitalWrite(greenLED, LOW);
          digitalWrite(redLED, HIGH); // Red LED for high pressure
        } else if (pressure > 1000) {
          digitalWrite(blueLED, LOW);digitalWrite(redLED, LOW);
          digitalWrite(greenLED, HIGH); // Green LED for normalpressure
        } else {
          digitalWrite(redLED, LOW); digitalWrite(greenLED, LOW);
          digitalWrite(blueLED, HIGH);}// Blue LED for low pressure
```

- Ensures only one LED is on at a time.
- Turns off all LEDs before activating the desired one.
- Uses if else statements to handle different colour indications.

CONCLUSION

Key Learnings

- ✓ Room Climate Monitoring: Measures temperature, humidity, and pressure, displaying data on an LCD with LED indicators.
- ✓ User-Friendly Interface: Keypad allows users to select which reading to view easily.
- ✓ Real-Time Alerts: LEDs indicate if conditions are too high, normal, or too low.

Future Improvements

- ❑ Smart Home Automation: Integrate with HVAC for automatic climate control.
- ❑ Greenhouses and Labs: Monitor and maintain optimal environmental conditions.
- ❑ Warehouses: Track conditions to protect sensitive stored items.